

Ultrafast X-Ray Experiments

**Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21**

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1. Einleitung

2. Theoretische Grundlagen

2.1. Synchrotronradiation

2.1.1. Free Electron Laser (FEL)

2.1.2. Undulators

2.2. The investigated Molecule

2.2.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$

2.2.2. Photophysics

2.3. Pump-Probe-Principle

2.3.1. The Pump-Pulse

2.3.2. The Probe-Pulse

3. Experimental Setup

3.1. Software: VLab

3.2. Undulator and Beamline

3.3. Monochromator and Focusing Optics

monochromator focusing optics: mirrors, power slits, solid attenuators, refractive Lenses, X-ray eye

3.4. Spectrum Analyser

diffraction gratings

3.5. Time-arrival Detector

3.6. Sample Environment

3.7. Laser System

3.8. Von Hamos Spectrometer

Quelle Versuchsmappe:[1]

4. Versuchsdurchführung

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

X-ray energy $E_0 = 7400\text{eV}$

Alignment: size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

Fun Facts: molar mass: $M([Fe(bipy)_3]^{2+}) \approx M([Fe(bipy)_3]) = 524,409 \frac{g}{mol}$ concentration of the sample: $c = 0,380 \frac{mol}{L}$ total attenuation of X-rays: attenuation comparison plot flow speed of the jet: $v_{jet} = 11 \frac{m}{s}$

Laser: excitation wavelength: $\lambda_{pump} = 400nm$ (dazu alle fun facts mit MLCT Zuständen und so, und nicht overshooten lol) focussing, diameter of the focus Overfilling: beam sizes of the optical laser and the X-ray pulse relative to each other (plot beam diameter)

Spectrometer: Van Hamos Spectrometer vs Johann Spectrometer (plot of the two geometries) crystal selection: $Si(531)$ vs $Ge(531)$ (plot of the two crystal reflections)

4.2. Performing the Experiment with Data Acquisition

5. Auswertung der Messergebnisse

6. Diskussion der physikalischen Ergebnisse

7. Zusammenfassung

References

- [1] C. Bressle, “Ultrafast x-ray experiments virtual experiments at the x-ray laser european xfel.”

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